

Part X - Rheumatology and Immunology

Chapter 10D: Vasculitis, Polymyalgia Rheumatica, Giant Cell Arteritis, Behcet Syndrome, and Sarcoidosis

Medvale Internal Medicine - Original Study Textbook Segment

Orientation and chapter purpose

This chapter moves from joint-centered connective tissue disease into systemic inflammatory syndromes that injure blood vessels, granulomatous tissues, eyes, skin, lungs, kidneys, nerves, and the central nervous system. The practical challenge is that these illnesses often present as fever, pain, rash, renal dysfunction, pulmonary infiltrates, neuropathy, or constitutional symptoms rather than as a single swollen joint. The clinician must rapidly decide whether the pattern is benign and outpatient-manageable or whether it signals threatened vision, pulmonary hemorrhage, glomerulonephritis, bowel ischemia, digital ischemia, or nervous system disease. Those red flags justify urgent specialty consultation and empiric immunosuppression after infection has been reasonably addressed.

Vasculitis means inflammation of vessel walls. Vessel caliber largely determines the clinical pattern: large-vessel disease causes limb claudication, pulse deficits, blood pressure asymmetry, aortitis, aneurysm, and ischemic cranial symptoms; medium-vessel disease causes nodules, livedo, mononeuritis multiplex, mesenteric ischemia, renal infarcts, and aneurysms; small-vessel disease causes palpable purpura, glomerulonephritis, pulmonary capillaritis with hemoptysis, alveolar hemorrhage, sinus disease, and neuropathy. The most dangerous mistake is treating all vasculitis as one disease. The second most dangerous mistake is treating fever, rash, and high inflammatory markers with steroids without first considering bacteremia, endocarditis, meningococcemia, disseminated gonococcal infection, tuberculosis, hepatitis-associated vasculitis, and other infection mimics.

Graph-RAG anchor: Treat vasculitis as a parent class connected to vessel size, immune mechanism, organ injury pattern, biopsy target, serologic marker, infection mimic, and induction/maintenance treatment. A query about “hemoptysis plus hematuria” should activate pulmonary-renal syndrome, ANCA-associated vasculitis, anti-GBM disease, lupus nephritis, and infection mimics rather than only one named disease.

A first-pass framework for suspected vasculitis

The first-pass question is not “which antibody is positive?” but “what organ is currently threatened?” The presence of visual symptoms, jaw claudication, diplopia, stroke symptoms, hemoptysis, hypoxemia, rapidly rising creatinine, active urinary sediment, severe abdominal pain out of proportion to exam, testicular pain, mononeuritis multiplex, CNS deficits, or digital gangrene changes the tempo from routine outpatient workup to urgent evaluation and often inpatient treatment.

Vasculitis is also a pathology diagnosis when tissue can be safely obtained. Skin, kidney, temporal artery, nerve/muscle, lung, or other organ biopsy may demonstrate necrotizing inflammation, granulomas, immune complex deposition, pauci-immune glomerulonephritis, or giant cells. Serology is useful but cannot replace clinical localization. ANCA positivity supports granulomatosis with polyangiitis, microscopic polyangiitis, or eosinophilic granulomatosis with polyangiitis in the right phenotype; it does not diagnose vasculitis by itself. Complement consumption suggests immune complex disease, cryoglobulinemia, lupus, infection-associated vasculitis, or urticarial vasculitis. Normal complement with pauci-immune renal disease points toward ANCA-associated disease.

Table 10D-1. Initial algorithm for suspected vasculitis

Step	Clinical question	Why it matters
1. Stabilize	Is there vision loss, pulmonary hemorrhage, glomerulonephritis, CNS disease, bowel ischemia, or digital ischemia?	These findings require urgent treatment decisions; waiting for every confirmatory test can cause irreversible organ loss.
2. Rule out infection mimics	Could this be endocarditis, sepsis, meningococcemia, disseminated gonorrhea, hepatitis, TB, fungal disease, or HIV?	Immunosuppression can worsen infection; cultures, imaging, and targeted testing are often needed before high-dose therapy when feasible.
3. Localize vessel size and organs	Large vessel, medium vessel, small vessel, or variable vessel? Which organs define severity?	The vessel-size pattern narrows the differential and predicts complications.
4. Screen labs	CBC, CMP, urinalysis with microscopy, urine protein/creatinine, ESR/CRP, complements, ANCA, hepatitis serologies, cryoglobulins when appropriate.	Urine and creatinine are high-yield because renal vasculitis can be clinically silent until advanced.
5. Image and biopsy	CTA/MRA/PET for large vessels; chest CT for nodules/hemorrhage; biopsy accessible involved tissue.	Tissue and vascular imaging convert a syndrome into a more durable diagnosis.
6. Treat by severity	Non-severe cutaneous disease versus organ- or life-threatening disease.	Treatment ranges from NSAIDs/colchicine/topicals to pulse steroids plus rituximab, cyclophosphamide, or biologic therapy.

Vasculitis by vessel size

Vessel-size classification is useful because symptoms arise downstream from the caliber of vessel being inflamed. Large-vessel vasculitis includes giant cell arteritis and Takayasu arteritis. Medium-vessel vasculitis includes polyarteritis nodosa and Kawasaki disease. Small-vessel vasculitis includes ANCA-associated vasculitis and immune complex small-vessel disorders. Behcet syndrome is often described as variable-vessel vasculitis because it can affect arteries and veins of multiple sizes. Sarcoidosis is not primarily a vasculitis but overlaps clinically because granulomatous inflammation can involve lungs, lymph nodes, skin, eyes, heart, nervous system, and calcium metabolism.

Table 10D-2. Vessel-size map for systemic inflammatory disease

Category	Diseases	Classic clues	High-yield complications
Large vessel	Giant cell arteritis; Takayasu arteritis	Limb claudication, pulse deficits, bruits, blood pressure asymmetry, aortitis; cranial symptoms in GCA.	Vision loss, stroke, aortic aneurysm/dissection, large-artery stenosis.
Medium vessel	Polyarteritis nodosa; Kawasaki disease	Livedo reticularis, nodules, ulcers, mononeuritis multiplex, renal infarcts/HTN without glomerulonephritis, mesenteric ischemia.	Bowel ischemia/perforation, renal artery aneurysm, neuropathy, coronary aneurysm in Kawasaki.
Small vessel - ANCA	GPA; MPA; EGPA	Pulmonary-renal syndrome, sinus/ENT disease, asthma/eosinophilia in EGPA, palpable purpura, neuropathy.	Rapidly progressive GN, alveolar hemorrhage, pulmonary nodules/cavitations, mononeuritis multiplex.
Small vessel - immune complex	IgA vasculitis, cryoglobulinemic vasculitis, anti-GBM disease, hypocomplementemic urticarial vasculitis, infection-associated vasculitis	Palpable purpura, arthralgia, abdominal pain, nephritis; low complement in many immune complex disorders.	Nephritis, GI bleeding, pulmonary hemorrhage in anti-GBM, neuropathy in cryoglobulinemia.
Variable vessel	Behcet syndrome	Recurrent oral ulcers plus genital ulcers, uveitis, skin lesions, pathergy, venous thrombosis, arterial aneurysm.	Vision-threatening uveitis, cerebral venous thrombosis, pulmonary artery aneurysm, bowel ulceration.
Granulomatous mimic/overlap	Sarcoidosis	Bilateral hilar adenopathy, lung infiltrates, erythema nodosum, uveitis, hypercalcemia, noncaseating granulomas.	Cardiac conduction disease, ventricular arrhythmia, neurosarcoidosis, pulmonary fibrosis.

Large-vessel vasculitis

Giant cell arteritis

Giant cell arteritis is a granulomatous vasculitis of older adults that preferentially involves extracranial branches of the carotid artery and may also involve the aorta and its major branches. It should be suspected in patients older than 50 years with new headache, scalp tenderness, jaw claudication, visual symptoms, diplopia, constitutional symptoms, elevated inflammatory markers, or polymyalgia rheumatica features. The critical complication is irreversible vision loss from ischemic optic neuropathy or retinal ischemia. Because vision loss can become permanent quickly, suspected GCA with visual symptoms is treated immediately rather than waiting for biopsy confirmation.

The history often contains several clues that are more discriminating than headache alone. Jaw claudication is pain or fatigue in the masseter muscles during chewing and reflects ischemia of branches supplying the jaw. Scalp tenderness may be noticed when combing hair or resting the head on a pillow. Diplopia, amaurosis fugax, transient visual obscurations, or new monocular vision loss should be treated as an emergency. Polymyalgia rheumatica symptoms - bilateral shoulder and hip girdle aching with prominent morning stiffness - commonly coexist with GCA. ESR and CRP are usually elevated, but neither test is perfect; a normal inflammatory marker profile lowers probability but does not absolutely exclude disease when the clinical picture is compelling.

Diagnosis is clinical, supported by inflammatory markers, temporal artery ultrasound or vascular imaging where available, and temporal artery biopsy. Biopsy classically shows granulomatous inflammation, disruption of the internal elastic lamina, and giant cells, but skip lesions can create false negatives. Treatment should not be delayed for biopsy; histologic yield remains useful for a period after steroid initiation. Once treatment begins, symptoms and inflammatory markers often improve quickly, but relapse is common during tapering. Tocilizumab, an IL-6 receptor inhibitor, is used as a steroid-sparing agent in selected patients, especially relapsing disease or high steroid toxicity risk.

Table 10D-3. Giant cell arteritis: recognition and action points

Feature	GCA pattern	Clinical action
Age	Almost always age >50; peak in older adults.	New cranial symptoms in a younger patient should prompt alternative diagnoses, but do not ignore large-vessel symptoms.
Cranial ischemia	New temporal headache, scalp tenderness, jaw claudication, diplopia, amaurosis fugax, visual loss.	High-dose glucocorticoids immediately; ophthalmology/rheumatology involvement if vision symptoms.
PMR association	Bilateral shoulder/hip stiffness, pain, constitutional symptoms.	Screen PMR patients for GCA symptoms at every visit.

Feature	GCA pattern	Clinical action
Labs	ESR and CRP usually elevated; anemia and thrombocytosis may occur.	Markers support diagnosis and follow response, but they are not a substitute for symptoms.
Testing	Temporal artery biopsy, ultrasound halo sign, CTA/MRA/PET for extracranial large vessels.	Do not delay treatment for testing when vision is threatened.
Treatment	High-dose prednisone; IV methylprednisolone may be used for acute visual loss; tocilizumab for steroid-sparing in selected patients.	Add bone, glucose, blood pressure, infection, and GI risk mitigation during steroid exposure.

Red flag: New headache plus jaw claudication or visual symptoms in a patient older than 50 is giant cell arteritis until proven otherwise. Give glucocorticoids promptly and arrange confirmatory testing; do not wait for biopsy if vision is threatened.

Polymyalgia rheumatica

Polymyalgia rheumatica is an inflammatory syndrome of older adults characterized by bilateral shoulder and hip girdle pain, marked morning stiffness, functional difficulty rising from a chair or lifting the arms, and elevated inflammatory markers. Patients describe pain and stiffness rather than true muscle weakness. This distinction matters: in PMR, strength testing is generally limited by pain but should be normal when pain is overcome; in inflammatory myopathy, objective proximal weakness predominates and CK is usually elevated. PMR is closely linked to GCA, so every PMR visit should include screening for new headache, jaw claudication, scalp tenderness, diplopia, or visual symptoms.

PMR often responds dramatically to low-dose prednisone, which is diagnostically supportive but not perfectly specific. A poor response should trigger reconsideration of mimics such as elderly-onset rheumatoid arthritis, statin myopathy, hypothyroidism, malignancy, infection, rotator cuff disease, cervical/lumbar spinal stenosis, and inflammatory myopathy. Treatment requires a slow taper because relapse is common. Methotrexate or IL-6 blockade may be used in recurrent disease or steroid toxicity risk, but the foundational teaching point is that PMR uses lower steroid doses than GCA and does not itself cause blindness.

Table 10D-4. PMR versus GCA and common mimics

Disorder	Dominant symptom	Key labs	Danger point	Treatment logic
PMR	Bilateral shoulder/hip girdle aching and morning stiffness; no true weakness.	ESR/CRP elevated; CK normal.	Associated GCA can cause vision loss; PMR alone does not.	Low-dose prednisone with slow taper; monitor relapse and steroid toxicity.
GCA	New headache, jaw claudication, scalp tenderness, visual symptoms; often PMR symptoms.	ESR/CRP elevated; platelets may be high.	Irreversible vision loss and stroke.	High-dose steroids immediately; biopsy/imaging; consider tocilizumab.
Inflammatory myopathy	Objective proximal weakness more than pain.	CK/aldolase elevated; myositis antibodies may help.	Dysphagia, interstitial lung disease, malignancy association.	Steroids plus steroid-sparing immunosuppression; evaluate lungs and cancer risk.
Elderly-onset RA	Peripheral synovitis, MCP/PIP/wrist involvement, erosive risk.	RF/anti-CCP may be positive; ESR/CRP elevated.	Joint destruction if untreated.	DMARD-centered, usually methotrexate if appropriate.

Takayasu arteritis

Takayasu arteritis is a granulomatous large-vessel vasculitis of the aorta and major branches, classically affecting younger women. It may begin with systemic symptoms such as fever, malaise, arthralgia, and weight loss before progressing to vascular stenosis and ischemia. Clinical clues include limb claudication, reduced or absent pulses, blood pressure differences between arms, bruits over subclavian or carotid arteries, renovascular hypertension, and neurologic symptoms from carotid or vertebral involvement. Diagnosis relies heavily on vascular imaging such as CTA, MRA, ultrasound, or PET. Treatment uses glucocorticoids and steroid-sparing immunosuppression or biologic therapy in active disease; fixed stenoses may require revascularization after inflammation is controlled.

Medium-vessel vasculitis

Polyarteritis nodosa

Polyarteritis nodosa is a necrotizing vasculitis of medium-sized muscular arteries. It classically spares arterioles, capillaries, and venules, so glomerulonephritis and pulmonary capillaritis are not typical. This is a high-yield distinction from microscopic polyangiitis and other small-vessel pulmonary-renal syndromes. PAN causes tissue ischemia through arterial inflammation, aneurysm formation, stenosis, thrombosis, and infarction. Common manifestations include fever, weight loss, livedo reticularis, tender subcutaneous nodules, ulcers, mononeuritis multiplex, myalgias, arthralgias, hypertension from renal artery involvement, renal infarcts, abdominal pain after meals, mesenteric ischemia, testicular pain, and peripheral neuropathy.

A classic board-relevant association is hepatitis B infection. When hepatitis B-associated PAN is suspected, antiviral therapy and plasma exchange may be part of management rather than treating only with long courses of immunosuppression. Angiography may

show microaneurysms and irregular constrictions in renal, hepatic, or mesenteric vessels. Biopsy of involved tissue can demonstrate necrotizing arteritis. ANCA is usually negative, which helps separate PAN from ANCA-associated small-vessel vasculitis. Severe PAN is treated with glucocorticoids plus cyclophosphamide or other immunosuppressive strategies depending on severity and cause.

Table 10D-5. PAN versus small-vessel ANCA vasculitis

Finding	PAN	Microscopic polyangiitis / ANCA small-vessel disease
Vessel target	Medium muscular arteries.	Small vessels: capillaries, venules, arterioles; may involve small arteries.
Kidney pattern	Renal artery involvement, infarcts, hypertension; no primary glomerulonephritis.	Pauci-immune necrotizing crescentic glomerulonephritis with hematuria/RBC casts.
Lung pattern	Pulmonary capillaritis is not typical.	Pulmonary hemorrhage can occur.
Serology	ANCA usually negative; consider hepatitis B testing.	MPO-ANCA often positive in MPA; PR3/MPO patterns vary by disease.
Classic symptoms	Livedo, nodules, mononeuritis multiplex, mesenteric ischemia, testicular pain.	Pulmonary-renal syndrome, palpable purpura, neuropathy, constitutional symptoms.
Testing clue	Angiographic microaneurysms in medium vessels.	Urinalysis with active sediment; biopsy shows pauci-immune small-vessel disease.

Kawasaki disease

Kawasaki disease is an acute medium-vessel vasculitis of childhood with fever, mucocutaneous inflammation, conjunctivitis, rash, extremity changes, and cervical lymphadenopathy. Its major internal medicine relevance is coronary artery aneurysm, which can cause myocardial ischemia later in life. Treatment in children uses IVIG and aspirin to reduce coronary complications. In an adult review chapter, Kawasaki is less commonly encountered than GCA, PMR, PAN, ANCA vasculitis, Behcet syndrome, or sarcoidosis, but it anchors the concept that medium-vessel vasculitis can target coronary arteries.

Small-vessel vasculitis and pulmonary-renal syndromes

ANCA-associated vasculitis

The ANCA-associated vasculitides are necrotizing, pauci-immune small-vessel vasculitides. “Pauci-immune” means little or no immune complex deposition on immunofluorescence compared with lupus nephritis or IgA vasculitis. The main syndromes are granulomatosis with polyangiitis, microscopic polyangiitis, and eosinophilic granulomatosis with polyangiitis. They are unified by small-vessel inflammation and potential for rapidly progressive glomerulonephritis and pulmonary hemorrhage, but separated by ENT granulomatous disease, asthma/eosinophilia, and serologic patterns.

ANCA patterns are useful but imperfect. PR3-ANCA often corresponds to a cytoplasmic immunofluorescence pattern and is classically associated with granulomatosis with polyangiitis. MPO-ANCA often corresponds to a perinuclear pattern and is classically associated with microscopic polyangiitis and eosinophilic granulomatosis with polyangiitis. However, clinical phenotype is more important than memorizing a one-antibody-one-disease rule. ANCA can be negative in true disease and can be positive in infection, drug exposure, inflammatory bowel disease, and other inflammatory states.

Table 10D-6. ANCA-associated vasculitis comparison

Disease	Signature phenotype	ANCA pattern	Kidney/lung pattern	Treatment concept
GPA	Chronic sinusitis, otitis, nasal crusting/ulcers, saddle-nose deformity, subglottic stenosis, pulmonary nodules/cavitations.	Often PR3-ANCA/c-ANCA; may be MPO or ANCA-negative.	RPGN; pulmonary nodules or alveolar hemorrhage.	Severe disease: high-dose steroids plus rituximab or cyclophosphamide; maintenance often rituximab, azathioprine, methotrexate, or other steroid-sparing plan.
MPA	No granulomatous ENT disease; palpable purpura, neuropathy, constitutional symptoms.	Often MPO-ANCA/p-ANCA.	Pulmonary capillaritis and pauci-immune crescentic GN are classic.	Similar severe-disease induction with steroids plus rituximab or cyclophosphamide; monitor renal and pulmonary response.
EGPA	Asthma, allergic rhinitis/sinusitis, eosinophilia, neuropathy, pulmonary infiltrates; can involve heart.	MPO-ANCA in a subset; many are ANCA-negative.	Renal disease less common than MPA/GPA; lung infiltrates and asthma dominate.	Steroids; add immunosuppression/biologic based on severity, eosinophilic features, and organ involvement.

Pulmonary-renal emergency: Hemoptysis, falling hemoglobin, diffuse pulmonary infiltrates, rising creatinine, hematuria, proteinuria, or RBC casts should trigger urgent evaluation for pulmonary-renal syndrome. Consider ANCA vasculitis, anti-GBM disease, lupus, infection, and drug-induced disease. Do not dismiss mild hemoptysis in a patient with active urinary sediment.

Granulomatosis with polyangiitis

GPA is the ANCA vasculitis most associated with destructive upper airway disease and granulomatous inflammation. Patients may have chronic sinusitis, otitis media, hearing loss, epistaxis, nasal crusting, oral or nasal ulcers, saddle-nose deformity, subglottic

stenosis, cough, dyspnea, hemoptysis, pulmonary nodules, cavitations, and glomerulonephritis. Because sinus disease is common in the general population, the diagnostic signal comes from persistence, destructive features, lower respiratory involvement, renal findings, systemic inflammation, and ANCA or biopsy support. Untreated severe GPA can rapidly progress to renal failure or fatal pulmonary hemorrhage.

Microscopic polyangiitis

MPA is a small-vessel necrotizing vasculitis without the granulomatous ENT masses that define GPA. It commonly presents with constitutional symptoms, palpable purpura, mononeuritis multiplex, pulmonary capillaritis, and rapidly progressive glomerulonephritis. MPO-ANCA is common. A key clinical distinction from PAN is the presence of glomerulonephritis and pulmonary capillaritis in MPA, whereas classic PAN has medium-artery renal ischemia and usually no lung capillaritis.

Eosinophilic granulomatosis with polyangiitis

EGPA should be suspected when asthma, allergic rhinitis, chronic sinus disease, pulmonary infiltrates, eosinophilia, neuropathy, and systemic features coexist. The disease can be conceptualized in phases: allergic airway disease, eosinophilic tissue infiltration, and systemic vasculitis. Mononeuritis multiplex is common and often clinically prominent. Cardiac involvement is a major cause of morbidity and mortality, so chest pain, heart failure signs, troponin elevation, arrhythmias, or cardiomyopathy require careful evaluation. ANCA is positive in only a subset; therefore absence of ANCA does not exclude EGPA.

Immune-complex small-vessel vasculitis

Immune-complex vasculitis differs from pauci-immune ANCA vasculitis by deposition of immunoglobulin and complement in vessel walls. This mechanism often produces low complement levels and palpable purpura. Examples include IgA vasculitis, cryoglobulinemic vasculitis, hypocomplementemic urticarial vasculitis, infection-associated vasculitis, and lupus vasculitis. The skin may be the easiest tissue to biopsy, especially if a fresh purpuric lesion is sampled for routine histology and direct immunofluorescence.

Table 10D-7. Immune-complex small-vessel vasculitis

Disorder	Classic clues	Complement/serology	Major organ concerns
IgA vasculitis	Palpable purpura, arthralgia/arthritis, abdominal pain, renal involvement; often after infection.	IgA deposition; complement often normal.	Nephritis, GI bleeding/intussusception in children, persistent renal disease.
Cryoglobulinemic vasculitis	Palpable purpura, arthralgia, weakness, neuropathy, renal disease; hepatitis C association.	Low C4 is classic; cryoglobulins; RF may be positive.	Membranoproliferative GN, neuropathy, skin ulcers, systemic disease.
Hypocomplementemic urticarial vasculitis	Urticarial lesions lasting >24 hours, burning/pain, residual hyperpigmentation, systemic symptoms.	Low complement; anti-C1q may be present.	Renal disease, pulmonary disease, ocular inflammation.
Infection-associated vasculitis	Purpura with bacteremia, endocarditis, hepatitis, HIV, or other infection signs.	Complement may be low; cultures/serologies guide diagnosis.	Treating with steroids alone can be harmful; control infection.
Lupus vasculitis	Occurs with other SLE features: nephritis, cytopenias, rash, serositis, anti-dsDNA activity.	Low C3/C4, anti-dsDNA activity.	Nephritis, CNS disease, digital ischemia, skin ulcers.

Variable-vessel vasculitis: Behcet syndrome

Behcet syndrome is a relapsing inflammatory disorder that can affect mucosa, skin, eyes, joints, veins, arteries, gastrointestinal tract, and nervous system. It is often taught through the triad of recurrent oral ulcers, genital ulcers, and uveitis, but this triad is incomplete. Oral aphthae are common and often precede diagnosis by years. Genital ulcers tend to scar. Ocular disease may include anterior uveitis, posterior uveitis, or retinal vasculitis and can threaten vision. Skin findings include erythema nodosum-like lesions, papulopustular lesions, acneiform lesions, and pathergy, in which minor trauma triggers exaggerated inflammatory papules or pustules.

The vascular behavior of Behcet is unusual because it can cause both venous thrombosis and arterial aneurysm. Venous thrombosis may involve deep veins, vena cava, cerebral venous sinuses, or hepatic veins. The thrombosis is inflammation-driven and adherent, so embolism is less central than in ordinary hypercoagulability; immunosuppression is a major treatment principle for vascular Behcet. Arterial disease can cause pulmonary artery aneurysms, which may present with hemoptysis and can be fatal. Gastrointestinal Behcet can mimic inflammatory bowel disease with ulceration, abdominal pain, bleeding, or perforation. Neurologic Behcet can involve brainstem, meninges, or cerebral venous sinuses.

Table 10D-8. Behcet syndrome by organ domain

Domain	Behcet clues	Treatment concept
Mucocutaneous	Recurrent oral aphthae, genital ulcers with scarring, erythema nodosum-like lesions, papulopustular lesions, pathergy.	Topical steroids for mild lesions; colchicine often first-line for recurrent mucocutaneous/joint disease; apremilast, azathioprine, TNF inhibitors, or other agents for refractory disease.

Domain	Behcet clues	Treatment concept
Ocular	Anterior/posterior uveitis, retinal vasculitis, eye pain, photophobia, floaters, vision change.	Urgent ophthalmology; systemic immunosuppression for posterior segment or vision-threatening disease; avoid treating serious eye disease with topical therapy alone.
Vascular	DVT, vena cava thrombosis, cerebral venous sinus thrombosis, arterial aneurysm, pulmonary artery aneurysm with hemoptysis.	Immunosuppression is central; anticoagulation decisions require expert input, especially if aneurysm or bleeding risk exists.
Neurologic	Meningoencephalitis, brainstem signs, cerebral venous sinus thrombosis, headaches with neurologic deficits.	High-dose steroids plus immunosuppressive therapy for severe parenchymal disease; evaluate thrombosis versus inflammation.
GI	Ileocecal ulcers, abdominal pain, bleeding, perforation; can mimic Crohn disease.	Severity-based immunosuppression/biologics; surgical evaluation for perforation or uncontrolled bleeding.

Behcet red flags: Eye symptoms, hemoptysis, neurologic deficits, severe abdominal pain, and major-vessel thrombosis are not “just aphthous ulcers.” These findings suggest organ-threatening Behcet disease and require urgent specialty evaluation.

Sarcoidosis

Sarcoidosis is a multisystem granulomatous disease characterized by noncaseating granulomas after exclusion of alternative causes such as tuberculosis, fungal infection, berylliosis, malignancy, and drug reactions. It commonly involves lungs and intrathoracic lymph nodes, but the disease is systemic. The classic chest imaging pattern is bilateral hilar lymphadenopathy, sometimes with parenchymal infiltrates. Symptoms may include cough, dyspnea, chest discomfort, fatigue, fever, night sweats, weight loss, arthralgia, erythema nodosum, uveitis, hypercalcemia, nephrolithiasis, facial nerve palsy, conduction disease, arrhythmias, cardiomyopathy, or neurologic deficits. Many cases are asymptomatic and discovered incidentally on chest imaging.

The diagnostic reasoning in sarcoidosis has three parts: compatible clinical-radiographic pattern, histologic evidence of noncaseating granulomas when tissue confirmation is needed, and exclusion of mimics. ACE levels are neither sensitive nor specific enough to diagnose sarcoidosis alone. Hypercalcemia and hypercalciuria occur because macrophages within granulomas can increase extrarenal conversion of vitamin D to active calcitriol. Eye, heart, and nervous system involvement matter because they can produce permanent organ injury even when pulmonary symptoms are mild.

Table 10D-9. Sarcoidosis organ manifestations

Organ system	Common manifestations	High-yield evaluation point
Lung/lymph nodes	Bilateral hilar adenopathy, cough, dyspnea, interstitial infiltrates, restrictive physiology.	Chest imaging plus pulmonary function tests; biopsy when diagnosis is uncertain or treatment decisions require confirmation.
Skin	Erythema nodosum, lupus pernio, plaques, papules, scar infiltration.	Skin lesions can provide accessible biopsy targets; lupus pernio suggests chronic disease.
Eye	Anterior or posterior uveitis, eye pain, photophobia, blurred vision.	Baseline and symptom-driven ophthalmology; untreated ocular disease can threaten vision.
Calcium/kidney	Hypercalcemia, hypercalciuria, nephrolithiasis, nephrocalcinosis.	Check calcium, creatinine, urine calcium when indicated; avoid excess vitamin D/calcium supplementation unless guided.
Cardiac	Conduction block, ventricular arrhythmias, cardiomyopathy, syncope, sudden death.	ECG screening; cardiac MRI/PET and cardiology evaluation for symptoms, abnormal ECG, or suspected involvement.
Nervous system	Facial nerve palsy, meningitis, neuropathy, pituitary/hypothalamic disease.	MRI/CSF and neurologic evaluation; exclude infection and malignancy.

Sarcoidosis treatment is based on organ threat rather than the mere presence of granulomas. Asymptomatic bilateral hilar adenopathy may be observed. Symptomatic pulmonary disease, ocular disease, neurologic disease, cardiac disease, severe hypercalcemia, disfiguring skin disease, or progressive organ dysfunction often requires systemic glucocorticoids and steroid-sparing therapy such as methotrexate, azathioprine, mycophenolate, leflunomide, or TNF inhibition in selected refractory cases. Because noncaseating granulomas are not unique to sarcoidosis, tuberculosis and fungal disease must be considered before immunosuppression, especially when epidemiology or imaging is atypical.

Table 10D-10. Sarcoidosis mimics to exclude

Sarcoidosis mimic	Why it matters	Clues against simple sarcoidosis
Tuberculosis	Steroids can worsen infection; granulomas may be caseating or noncaseating.	Exposure risk, cavitation, constitutional symptoms, positive TB testing, AFB culture/PCR.
Histoplasmosis/other fungi	Endemic mycoses can cause pulmonary nodules and adenopathy.	Travel/endemic exposure, fungal serology/antigen/culture, immunosuppression risk.
Lymphoma/malignancy	Can mimic adenopathy and systemic symptoms.	B symptoms, asymmetric bulky nodes, cytopenias, abnormal PET pattern requiring biopsy.
Berylliosis	Granulomatous lung disease from occupational exposure.	Aerospace, manufacturing, ceramics, electronics exposure; beryllium lymphocyte proliferation testing.
Drug-induced sarcoid-like reaction	Checkpoint inhibitors and other drugs can trigger granulomatous inflammation.	Temporal relation to immune therapy; oncology context.

Vasculitis laboratory interpretation

Inflammatory markers are supportive but nonspecific. ESR rises with age, anemia, pregnancy, kidney disease, and hypergammaglobulinemia, and it changes more slowly. CRP rises and falls more dynamically and is often more useful for acute inflammatory change. Very high ESR/CRP in an older adult with headache or shoulder/hip stiffness supports GCA/PMR, but infection and malignancy remain important. Conversely, normal inflammatory markers reduce probability but do not replace clinical judgment when the consequence is blindness, renal failure, or pulmonary hemorrhage.

Urinalysis is one of the most important tests in suspected systemic vasculitis. Hematuria, dysmorphic RBCs, RBC casts, and proteinuria suggest glomerulonephritis even when the patient has no urinary symptoms. Creatinine may lag behind active inflammation. A “normal-looking” patient with purpura and RBC casts is not low risk. In pulmonary-renal syndromes, the combination of urinalysis, creatinine trend, chest imaging, hemoglobin, oxygenation, ANCA, anti-GBM antibody, complements, and lupus serologies guides urgent therapy and biopsy decisions.

Table 10D-11. Lab interpretation in suspected vasculitis

Test	High-yield interpretation	Pitfall
ESR/CRP	Support systemic inflammation and track response in PMR/GCA and many vasculitides.	Nonspecific; infection, malignancy, obesity, trauma, and chronic disease can elevate markers.
CBC	Anemia/thrombocytosis can accompany inflammation; eosinophilia suggests EGPA or drug/allergic disease; leukopenia may suggest lupus/drug/viral disease.	Steroids can raise WBC count; cytopenias may point away from isolated vasculitis.
Creatinine + urinalysis	RBC casts, dysmorphic RBCs, hematuria, and proteinuria suggest glomerulonephritis.	Renal vasculitis can be asymptomatic; do not wait for flank pain or urinary complaints.
ANCA PR3/MPO	Supports ANCA-associated vasculitis in the right phenotype.	False positives occur; ANCA-negative disease exists; do not screen indiscriminately without compatible syndrome.
Complement C3/C4	Low complement supports immune complex disease, cryoglobulinemia, lupus, infection-associated vasculitis.	Normal complement does not rule out all immune complex disease.
Hepatitis B/C, HIV	Identify infection-associated vasculitis and treatment-altering causes such as HBV-PAN or HCV cryoglobulinemia.	Missing hepatitis testing can lead to wrong immunosuppression strategy.
Cryoglobulins	Consider with purpura, neuropathy, renal disease, low C4, hepatitis C risk.	Sample handling matters; false negatives occur if specimen is not kept warm before processing.
Biopsy	Provides tissue pattern: granulomatous, necrotizing, immune-complex, pauci-immune.	Wrong lesion or late scarred tissue can be nondiagnostic; coordinate biopsy target.

Treatment architecture for systemic vasculitis

Treatment is organized around severity and phase. Induction therapy controls active inflammation and prevents organ loss. Maintenance therapy prevents relapse while minimizing toxicity. Non-severe disease may be treated with topical therapy, NSAIDs, colchicine, low-dose glucocorticoids, methotrexate, azathioprine, or other steroid-sparing agents depending on disease. Organ- or life-threatening disease often requires high-dose glucocorticoids plus rituximab, cyclophosphamide, or disease-specific biologic therapy. Plasma exchange has a narrower role than historically used and is considered in selected severe pulmonary-renal or anti-GBM contexts rather than as automatic treatment for all ANCA vasculitis.

Before immunosuppression, clinicians should assess infection risk, vaccination status, latent tuberculosis or hepatitis risk when biologics are planned, pregnancy considerations, fertility preservation when cyclophosphamide is used, bone protection, Pneumocystis prophylaxis when high-risk regimens are used, gastric and glucose risks, and baseline organ monitoring. This matters because the treatment can be as dangerous as the disease if applied without prevention planning.

Table 10D-12. Immunosuppressive medication overview

Medication/class	Common role in this chapter	Major toxicities/monitoring
Glucocorticoids	Rapid control of GCA, PMR, severe vasculitis, sarcoidosis flares, Behcet organ disease.	Hyperglycemia, hypertension, infection, osteoporosis, avascular necrosis, mood/insomnia, myopathy, cataracts/glaucoma; taper carefully.
Rituximab	Induction/maintenance for ANCA-associated vasculitis; some refractory autoimmune disease.	HBV reactivation, hypogammaglobulinemia, infusion reactions, infection; check hepatitis B and immunoglobulins when appropriate.
Cyclophosphamide	Severe organ-threatening vasculitis when potent induction is needed.	Myelosuppression, infection, hemorrhagic cystitis, infertility, bladder cancer, malignancy risk; CBC/urinalysis and fertility counseling.
Methotrexate	Steroid-sparing for non-severe vasculitis/PMR/sarcoidosis; RA/PsA overlap.	Hepatotoxicity, cytopenias, mucositis, pneumonitis, teratogenicity; folic acid, CBC/CMP monitoring.
Azathioprine	Maintenance therapy in vasculitis; sarcoidosis/Behcet steroid-sparing.	Myelosuppression, hepatotoxicity, infection; TPMT/NUDT15 consideration, CBC/CMP monitoring.
Mycophenolate	Alternative steroid-sparing agent in selected vasculitis/sarcoid/Behcet contexts.	GI upset, cytopenias, infection, teratogenicity; pregnancy prevention and CBC monitoring.
Tocilizumab	Steroid-sparing in GCA; sometimes refractory PMR.	Infection, neutropenia, transaminitis, lipid elevation, bowel perforation risk; CRP may be suppressed despite infection.

Medication/class	Common role in this chapter	Major toxicities/monitoring
TNF inhibitors	Behcet ocular/vascular/GI disease, refractory sarcoidosis in selected cases.	TB/fungal infection, HBV reactivation, demyelination/heart failure cautions; screen for latent infection.
Colchicine	Behcet mucocutaneous/joint disease; some neutrophilic inflammatory patterns.	GI intolerance, cytopenias/myopathy especially with renal disease or interacting drugs.

Diagnostic algorithms

Algorithm 1: new purpura with systemic symptoms

1. Confirm morphology: palpable purpura suggests small-vessel vasculitis; retiform purpura or necrosis suggests vessel occlusion, medium-vessel disease, antiphospholipid syndrome, emboli, calciphylaxis, or severe infection.
2. Assess severity immediately: fever, hypotension, meningismus, necrosis, severe abdominal pain, neuropathy, hematuria/RBC casts, creatinine rise, hemoptysis, hypoxemia, or altered mental status.
3. Screen for infection and mimics: blood cultures if febrile, endocarditis evaluation when murmur/emboli/risk factors exist, hepatitis/HIV testing when relevant, medication and cocaine/levamisole exposure history.
4. Order first-line inflammatory/renal tests: CBC, CMP, ESR/CRP, urinalysis with microscopy, urine protein/creatinine, complements, ANCA, ANA/dsDNA if lupus features, cryoglobulins if low C4/purpura/neuropathy/HCV risk.
5. Biopsy a fresh skin lesion for histology and direct immunofluorescence when diagnosis is uncertain or systemic disease is present.
6. Treat based on organ involvement rather than rash appearance alone; isolated cutaneous disease differs from nephritis, pulmonary hemorrhage, neuropathy, or bowel ischemia.

Algorithm 2: pulmonary-renal syndrome

1. Recognize the syndrome: hemoptysis may be absent; suspect alveolar hemorrhage with dyspnea, hypoxemia, diffuse infiltrates, falling hemoglobin, or bronchoscopy showing progressively bloody lavage plus glomerulonephritis markers.
2. Immediate tests: CBC, CMP, urinalysis with microscopy, urine protein/creatinine, ESR/CRP, ANCA PR3/MPO, anti-GBM antibody, ANA, anti-dsDNA, complements, coagulation studies, chest imaging, infection cultures/testing.
3. Consult nephrology/rheumatology/pulmonology early. Kidney biopsy is often the highest-yield tissue test if safe because it distinguishes pauci-immune GN, anti-GBM disease, immune-complex GN, and other causes.
4. If severe disease is likely, treatment often begins before every result returns: high-dose steroids plus disease-specific induction therapy. Cover infection when clinically plausible until cultures and imaging clarify.
5. Continue monitoring oxygenation, hemoglobin, creatinine, urine sediment, and treatment complications; pulmonary hemorrhage and RPGN can evolve quickly.

Algorithm 3: older adult with shoulder/hip stiffness

1. Ask whether symptoms are pain/stiffness or true weakness. PMR causes pain-limited function without objective weakness; myositis causes true proximal weakness and high CK.
2. Check ESR/CRP, CBC, CMP, CK, TSH if unclear, and evaluate for peripheral synovitis, malignancy/infection symptoms, and medication causes.
3. Screen for GCA symptoms at the same visit: new headache, scalp tenderness, jaw claudication, diplopia, amaurosis fugax, or vision loss.
4. If PMR without GCA red flags is likely, low-dose prednisone usually produces rapid improvement; taper slowly and reassess if response is incomplete.
5. If any GCA red flag exists, escalate to urgent GCA evaluation and high-dose steroid strategy rather than PMR-dose treatment.

Board-style discriminators and memory anchors

Table 10D-13. Rapid discriminators

Presentation	Think first	Reasoning shortcut
Age >50, new headache, jaw claudication, visual symptoms	Giant cell arteritis	Cranial ischemia plus age threshold equals emergency steroids.
Age >50, bilateral shoulder/hip stiffness, high ESR/CRP, normal CK	Polymyalgia rheumatica	Pain/stiffness without weakness; screen for GCA.
Young woman with absent pulses, arm BP discrepancy, bruits	Takayasu arteritis	Large-vessel stenosis in younger patient.

Presentation	Think first	Reasoning shortcut
Livedo, nodules, mononeuritis multiplex, abdominal pain, HBV, renal HTN without GN	Polyarteritis nodosa	Medium-vessel ischemia; lung and glomeruli usually spared.
Sinusitis, pulmonary nodules/cavities, RPGN	GPA	ENT + lung + kidney granulomatous ANCA disease.
Pulmonary hemorrhage + RPGN without granulomatous ENT disease	MPA	Small-vessel pulmonary-renal syndrome.
Asthma + eosinophilia + neuropathy + pulmonary infiltrates	EGPA	Allergic/eosinophilic disease progressing to vasculitis.
Palpable purpura + abdominal pain + arthralgia + nephritis	IgA vasculitis	IgA immune-complex small-vessel disease.
Purpura + neuropathy + low C4 + hepatitis C risk	Cryoglobulinemic vasculitis	Immune complexes precipitate; C4 is often low.
Recurrent oral/genital ulcers + uveitis + thrombosis/aneurysm	Behcet syndrome	Variable-vessel disease with mucosal and ocular anchors.
Bilateral hilar adenopathy + erythema nodosum/uveitis/hypercalcemia	Sarcoidosis	Noncaseating granulomatous disease after excluding mimics.

Integrated cases

Case 1: An older woman develops abrupt bilateral shoulder and hip stiffness with difficulty getting dressed and rising from a chair. ESR and CRP are elevated, CK is normal, and strength is intact when pain is overcome. This is a PMR pattern. The next critical question is not only how to treat pain, but whether she has GCA symptoms. If she denies headache, jaw claudication, scalp tenderness, diplopia, and visual symptoms, low-dose prednisone with taper is reasonable. If she reports transient monocular vision loss, the case becomes GCA until proven otherwise and requires urgent high-dose steroid treatment and confirmatory evaluation.

Case 2: A man has chronic sinusitis, otitis, cough, cavitory pulmonary nodules, hematuria, RBC casts, and rising creatinine. PR3-ANCA is positive. The pattern is GPA with renal involvement. The red flag is not the sinusitis; it is the active urinary sediment and pulmonary lesions. Induction therapy is considered organ-threatening, and kidney biopsy may confirm pauci-immune crescentic glomerulonephritis. Infection, especially TB or fungal disease in cavitory lung lesions, must be considered, but the pulmonary-renal pattern requires urgent action.

Case 3: A patient with recurrent oral aphthae and scarring genital ulcers develops painful red eye, photophobia, and blurred vision. Behcet syndrome is likely, but the key management point is that ocular involvement can threaten vision. Topical therapy for mouth ulcers is inadequate for posterior uveitis or retinal vasculitis. Ophthalmology and systemic immunosuppression are central.

Case 4: A patient has bilateral hilar lymphadenopathy, ankle arthritis, erythema nodosum, and uveitis. Sarcoidosis is likely, but the clinician must still exclude infection and malignancy when the presentation is atypical. If symptoms are mild and organ function is preserved, observation may be appropriate. If there is cardiac conduction disease, neurosarcoidosis, or vision-threatening ocular disease, systemic treatment becomes more urgent.

Chapter summary

- Classify vasculitis by vessel size, but treat based on organ threat and severity.
- GCA is a vision-threatening emergency in older adults with new cranial ischemic symptoms; start steroids immediately when suspected.
- PMR causes pain and stiffness without true weakness and is treated with lower steroid doses than GCA; always screen for GCA.
- PAN is medium-vessel disease with renal ischemia, mesenteric ischemia, mononeuritis multiplex, livedo, and hepatitis B association; it usually lacks glomerulonephritis and pulmonary capillaritis.
- ANCA-associated vasculitis causes pauci-immune small-vessel disease; pulmonary-renal syndrome requires urgent evaluation and treatment.
- Immune-complex vasculitis often has low complement and palpable purpura; infection-associated disease must be excluded before immunosuppression.
- Behcet syndrome is variable-vessel disease anchored by oral/genital ulcers but dangerous because of uveitis, thrombosis, aneurysm, CNS disease, and GI ulceration.
- Sarcoidosis is noncaseating granulomatous disease after exclusion of mimics; cardiac, neurologic, ocular, and calcium-related disease determine urgency.
- Urinalysis with microscopy is a high-yield, low-cost screen for occult glomerulonephritis in systemic inflammation.
- Immunosuppressive therapy requires infection screening, vaccination planning, bone protection, metabolic monitoring, and steroid-sparing strategy whenever feasible.

Selected clinical references used for synthesis

- American College of Rheumatology/Vasculitis Foundation guideline frameworks for ANCA-associated vasculitis, giant cell arteritis, and Takayasu arteritis.
- EULAR recommendations for Behcet syndrome management and organ-based treatment principles.
- Merck Manual Professional clinical summaries on vasculitis, giant cell arteritis, polymyalgia rheumatica, and sarcoidosis.
- NIAMS educational materials on polymyalgia rheumatica and giant cell arteritis.
- Current clinical review standards for sarcoidosis diagnosis emphasizing compatible syndrome, granulomatous pathology when needed, and exclusion of mimics.